P. Brown

SIXTH ANNUAL IN-HOUSE CONVENTION 22

DEPARTMENT OF PSYCHOLOGY

DALHOUSIE UNIVERSITY

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Sixth Annual In-House Convention

Department of Psychology

Dalhousie University

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Schedule

9:00 - 9:15	COFFEE

Session I. Chairman: J. Mates

9:15 - 9:30	1.	R. Klein
9:30 - 9:45	2.	M. Yoon
9:45 - 10:00	3.	D. Quine
10:00 - 10:15	4.	C. Berryman and J. Barresi
10:15 - 10:30	5.	J. Willner and R. Brown

10:30 - 11:00	COFFEE

Session II. Chairman: E. Hansen

11:00 - 11:15	6.	G. Goddard et al.
11:15 - 11:30	7.	G. Starratt
11:30 - 11:45	8.	K. Beverley and D. Regan
11:45 - 12:00	9.	L. White and J. Ryon
12:00 - 12:15	10.	C. Shaw, J. Mates and B. Rusak
12:15 - 12:30	11.	R. Brown and J. Walker

12:30 - 2.00	LUNCH AND POSTER SESSION

Session III: Chairman: R. Brown

2:00 - 2:15	12.	J. Ryon
2:15 - 2:30	13.	J. Barresi and B. MacCallum
2:30 - 2:45	14.	W.J. Jacobs, J. Hill-Flewelling and V. LoLordo
2:45 - 3:00	15.	J. Mates and P. Paton
3:00 - 3:15	16.	P. Dodd
3:15 - 3:30	17.	G. Eskes

3:30 - 4:00	COFFEE

Session IV. Chairwoman: J. Raymond

4:00 - 4:15	18.	K. Shapiro
4:15 - 4:30	19.	D. Regan and K. Beverley
4:30 - 4:45	20.	J. Fentress
4:45 - 5:00	21.	G. Hollway and M. Ozier
5:00 - 5:15	22.	B. Rusak

5:30 -	BAR OPEN

ABSTRACTS

1. R. Klein. Does Saccadic Suppression Have an Attentional Component?

The allocation of attention to a particular task will reduce the likelihood that nonattended inputs will be remembered or have access to any but their most habitual associations. While many processes are automatic in the sense that they do not require attention, others will suffer if attention is not available. It has been demonstrated that manual aimed movements and manual pursuit tracking usually require attention. The similarity of saccadic and pursuit eye movements to these manual movements suggests that they too might require attention. Subjective experience, on the other hand, suggests that eye movement control is usually reflexive or unconscious. It is proposed that whenever saccades must conform to spatial and temporal restrictions (as in most experiments where E tells S where and when to look, and in many skills, e.g. reading) their initiation and execution may not be automatic. This proposal predicts that the perception of stimuli occurring during a saccade will be suppressed (in relation to a no movement control) because attention will not be available to deal with them. It is interesting to note that suppression of visual perception prior to and during saccades has been well documented, but almost without exception this "saccadic suppression" phenomenon has been interpreted as a specifically visual effect. No studies that I am aware of have tested for the suppression of nonvisual stimuli, which if obtained would indicate an attentional contribution to saccadic suppression. The design and results of just such an experiment will be described.

2. M. Yoon. The Brain Through a Glass Window.

3. D. B. Quine. Groaning Mountains and Grunting Storms: The Challenge of Outdoor Research.

The laboratory demonstration of new senses in animals is most exciting when it is supported by field demonstrations that these new senses have an adaptive role in the life of the animal. My recent discovery that homing pigeons can detect and discriminate extremely low tones (infrasounds) raises some exciting questions about the function of this sensitivity. The difficulties associated with field studies of infrasound are representative of the challenges of working with any new sensory process. Many of these problems are due to the very properties of infrasound which make it a potentially useful environmental factor to many animals.

4. C. Berryman and J. Barresi. The Acquisition of Tacit and Explicit Knowledge of Contingent Probabilistic Structure.

In a dual processing paradigm subjects are presented with cues (2 digit numbers) shortly followed by one of four letters, where the individual letter probabilities depend on the particular cue. The subject's major task is to respond as rapidly as possible to each letter in a two choice decision procedure. His minor task is to attend to the relationship between the numbers and letters and at the end of each block to estimate their probabilistic structure. Evidence from an initial experiment with this paradigm suggests that the acquisition of tacit knowledge (reaction time) and explicit knowledge (verbal awareness) of probabilistic structure may involve partially independent processes.

5. J. Willner and R. Brown. Development of Odour Preferences in Rat Pups.

Two experiments examined the proposition that rat pups' preference for maternal odour reflects simple familiarity with the dam's scent. It was found that odour preexposure could affect pups' odour preferences; however, the data also indicate that a simple familiarity model does not provide a complete account of maternal odour preferences.

6. G. V. Goddard, T.V.P. Bliss, H. A. Robertson and R. S. Sutherland. Evidence to Support Crow's Reinforcement Hypothesis that Biologically Significant Behavior is Associated with Activity of the Locus Coeruleus and that the Consequent Release of Noradrenaline throughout the Forebrain has the Effect of Strengthening the Efficacy of Recently Active Synapses.

Rats were depleted of hippocampal noradrenaline by systemic injection of reserpine or by 6-OH-dopamine destruction of the dorsal noradrenergic bundle. Verification was by radioenzymatic liquid scintillation count. Brief high frequency electrical stimulation of perforant path fibers was used to cause short-term and long-term potentiation of the synapses onto granule cells as measured by the field potential response to test shocks in the perforant path. Short term potentiation of the field excitatory post-synaptic potential (EPSP) was unaffected by noradrenaline depletion whereas long-term potentiation of the EPSP was reduced by more than 50%.

7. G. Starratt. Mental Size Scaling.

Previous work by Burdeson and Larsen (1975), has demonstrated a phenomenon in which reaction time to determine if a pair of figures has the same shape is a linear function of the size ratio between the two figures. The functions for same and different data are parallel. Research by Klein (1979), also showed linear functions for "same" data but parallel functions were not reported for "different" data. Klein's data suggested that "complexity" of the stimuli could be a factor affecting the data. The first of two experiments run tested this complexity hypotheses. The second experiment was designed with the intention of resolving the discrepancies in the Klein and Bundeson "different" data. Results from the first experiment lend some support to Klein's complexity hypothesis and data from the second experiment indicate the differences in the Klein and Burdeson studies for the different data was due to the type of "different" patterns used.

8. K. I. Beverley and D. Regan. Visual Sensitivity to the Shape and Size of a Moving Object.

When a nonrotating solid object moves in three dimensions, its shape severely restricts the possible relationships between the velocities of the vertical and horizontal edges of its retinal image. The human visual system may contain detectors which achieve sensitivity to the shape of an object by utilizing these geometrically-determined velocity relationships.

9. L. White and J. Ryon. Dropped Ducks.

Would your newborn offspring benefit from a fall off a ten-storey building? An equivalent drop seems to increase the chance that a newly hatched captive wood duckling (Aix sponsa) will survive. In a pilot study last May we found that the activity of wood ducklings dropped from a height of at least a metre also differed from those gently placed into their brooder. This year we plan to investigate this phenomenon further, under controlled conditions.

10. C. Shaw, J. Mates and B. Rusak. The Psychology of Brain Slices.

Several researchers in the Dalhousie Psychology Department have become interested in using brain slices in their work. B. Rusak plans to record spontaneous neural activity in the hamster suprachiasmatic nuclei and to study the effects of pharmacological manipulations in relation to the circadian phase during which recordings are made. C. Shaw plans to examine the neural circuitry and pharmacology of visual cortex in cats. J. Mates wants to test the notion that the inhibitory feedback network containing hippocampal pyramidal cells acts laterally between adjacent lamellae of the hippocampus.

11. R. Brown and J. Walker. Play in Hospitalized and Non-Hospitalized Children.

Are hospitalized children under stress? Does the existence of a playschool in the hospital reduce this stress? In this initial study, the play behaviour of hospitalized children was compared with that of children in a daycare center. Differences in levels of play, social behaviour and "emotional" displays will be discussed.

POSTER PRESENTATIONS

A. D. Moore. Scanning Electron Microscope Study of the Eye Disc in the Larval Fly.

Larval diptera show no external indication of their adult features. However, early in development, cells within the body are sequestered away to form imaginal discs, which transform into such adult structures as genitalia, legs, wings, eyes and antennae. A scanning electron microscope study was conducted on one of these discs, the eye disc , in the 3rd instar larva of the housefly, Musca domestica.

B. L. Smith. Hemispheric Asymmetry in Part-Whole Judgements.

The work presented here is an attempt to obtain a test sensitive to right hemisphere processes. Experiments with stick figures, extracted from the 16 segments obtained from a 3 x 3 matrix, appear promising. A three-segment part, either contained within a foveally-presented six-segment figure or not, was briefly exposed in the left or right visual field. Decisions (part included in the figure or not) were made quicker for "easy" parts (e.g. triangles) in the right visual field, and for "difficult" parts (unconnected and scattered) in the left. These data suggest a method for evaluating perceptual and memory representations in the two hemispheres.

C. N. Bennett. Some Behavioral Correlates of Mate-Choice in the Coyote (Canis latrans).

D. J. Raymond. Abnormal Grating Adaptation in Multiple Sclerosis.

Reduced contrast sensitivity is often found in individuals with multiple sclerosis (MS). I investigated this abnormality of contrast vision by measuring changes in contrast sensitivity after adaptation to a high contrast sine wave grating pattern in MS patients and normal control subjects. I found that after viewing a 70% contrast adapting pattern for 1 minute, 15 out of 17 MS patients showed contrast sensitivity changes, or adaptation effects, smaller than 2.0 standard deviations below the mean adaptation effect for the normal group with at least one eye. In normals, the size of the adaptation effect increases both as the contrast of the adapting pattern is increased and as the duration of exposure to the adapting pattern is increased. Although these relationships also hold for the MS patients studied, the shapes of both the contrast and duration functions are different for the MS group.

When the contrast of the adapting pattern is low but still visible to the subject (less than 1.0 l.u. greater than the unadapted log contrast threshold), an adaptation effect can always be observed in normal subjects. However, in some MS patients adapting patterns of much higher contrasts (up to 1.0 l.u greater than the log unadapted contrast threshold) are insufficient to induce adaptation effects. Similarly, 20 to 60 seconds of exposure to a 70% contrast adapting pattern are enough to produce near-maximum adaptation effects in normal subjects. MS patients require exposures of 180 to 300 seconds before adaptation effects similar in size to those found in normals can be observed.

These differences between MS patients and normal subjects cannot be explained by reduced contrast sensitivity and indicate that the neural, probably cortical, mechanisms mediating visual contrast adaptation may be considerably altered by MS.

E. I. Meinertzhagen and A. Fröhlich. Synaptogenesis in the Fly's Visual System.

The ultrastructure and composition of fly photoreceptor synapse populations has been examined by serial electron microscopy of developing animals. Adult synapses are tetrads with four postsynaptic elements. At younger stages dyad and triad synapses (with two and three postsynaptic elements, respectively) coexist alongside tetrads, those of older animals having an average more postsynaptic processes per synapse. It is suggested that individual synapses assemble piecemeal, element by element.

F. D. Quine. 8 Octaves Below Human Hearing: Pigeons Detect Infrasound.

Homing pigeons can detect sounds at frequencies as low as 0.05 Hz. This poster will summarize the procedures used in measuring the infrasound sensitivity of homing pigeons. Frequency discrimination thresholds at these frequencies will be discussed in the context of what adaptive value these unusual sensory abilities may have to the pigeon. Possible uses include navigation, weather prediction and flight control. A comparison will be made with human low frequency hearing thresholds.

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12. J. Ryon. Aspects of Dominance Behavior in Groups of Sibling Coyotes.

end 2.20

Two unrelated groups of sexually mature sibling coyotes with three males and two females each were observed when each group was a) housed together, b) divided by sex into four groups and c) reunited after three months. Two dominance hierarchies (male and female) in each group were determined by social display. The effect of opposite-sex animals on 1) prosocial and 2) agonistic interactions, 3) scent-marking, 4) some aspects of howling and 5) spatial containment of low-ranking animals was studied. Same-sex aggression was higher in females than in males. The presence of males correlated with a striking increase in levels of aggression between females. The rate of scent-marking and howling was greater for the dominant females than for any other individual of either group. The only incident of spatial containment occurred among females.

13. J. Barresi and B. MacCallum. From Percept to Concept: Parsing Principles for Perceptual Categories.

2.35

When subjects are asked to partition random 9-dot patterns into their "natural" parts, most of the subjects' partitions for a particular pattern can be fit into a single nested hierarchy of parts. Such hierarchies often differ even among quite "similar" patterns. How then do we form abstract concepts to represent a category of such related patterns? Perhaps, with experience, related patterns assimilate to a single nested hierarchy of parts which comes to define the concept. Evidence will be presented which supports this hypothesis.

14. W. J. Jacobs, J. Hill-Flewelling, and V. LoLordo. More Quantitative Properties of Anxiety.

2.50

Three experiments investigated the effects of a CER procedure (Estes and Skinner, 1941) upon appetitively-motivated wheel running. Rats received the onset of a tone as a conditional stimulus and the presentation of a brief electric shock as the unconditional stimulus. Experiment 1 found that, unlike the well-documented suppression of an appetitively-motivated lever-press response, the current preparation produced acceleration of appetitively-motivated wheel running in the presence of the conditioned stimulus. Facilitation was not observed in the Random CS or No CS control conditions. Experiment 2 demonstrated that conditioned facilitation is a function of the intensity of the UCS: the more intense the UCS, the greater the facilitation. In Experiment 3, each rat was trained both to run in a wheel and to press a lever for food reinforcement. It was found that the rats increased their rate of wheel running, but suppressed their rate of lever pressing in the presence of the conditional stimulus. The results are discussed in the context of Dickinson and Pearce's recent (1977) model of conditioned suppression.

15. J. Mates and P. Paton. <u>The Malheur Swallows: A Simple Field Experiment Related to Flight Mechanisms.</u>

308 Lifting the wings of captive Cliff Swallows, *Petrochelidon pyrrhonota*, we triggered a fluttering of the body and tail. Onset and termination of the flutter occurred at fixed wing angles. Autonomous action of mechanisms underlying the flutter may drive the fluttering of birds voluntarily holding their wings up and back from their bodies.

16. P. Dodd. <u>The Response-Reinforcer Relation: A Pigeon's Eye View.</u>

324 Each presentation of a red response key ended with a food presentation either following two responses separated by at least 10 sec (a DRL schedule) or following a 10sec response-free period (a DRO schedule). A white side key was shown next and ended with response-dependent food following one schedule and a timeout following the other. Three pigeons responded differentially following the DRL and DRO schedules, but not simply according to the schedule duration, numbers of responses or even the time between a response and its consequence.

17. G. Eskes. <u>Functional Aspects and Neural Control of Daily Cycles of Male Sexual Performance.</u>

343 The circadian and neural mechanisms important for the functional organization of reproductive behavior in the male golden hamster were examined. It was found that males show a diurnal rhythm of sexual behavior; latencies to copulate and ejaculate were shorter in the dark phase, a time that coincides with the female period of sexual receptivity. The integrity of the suprachiastmatic nuclei of the anterior hypothalamus appears necessary for this rhythm. SCN lesions eliminated the rhythm in performance without interferring with the ability of the animal to copulate. In contrast, the medial preoptic area (MPOA) is essential for normal copulatory performance; in those animals that continued to copulate, after MPOA lesions the circadian modulation of sexual behavior remained intact. The regulation of sexual behavior by these structures constitutes a useful model for the interaction of circadian and other neural systems in the control of behavioral expression.

The significance of this rhythm is being tested by examining the ability of arrhythmic SCN-lesioned males or males in different circadian phases to copulate in a mating situation in which they must compete for access to a single receptive female. If the rhythm of copulatory readiness has adaptive value, then arrhythmic males or males tested in their light phase may be at a disadvantage relative to males tested in their dark phase in terms of reproductive success and fitness.

18. K. Shapiro. Selective Associations.

There have been a number of demonstrations in recent years that not all stimuli are associable with all reinforcers. One such selective association is evidenced when a compound auditory/visual (A-V) stimulus is paired with an appetitive reinforcer. After the compound A-V stimulus has acquired a high level of conditioned responding, the elements comprising this compound are tested for stimulus control. The visual element appears to overshadow or exert the dominant control over responding. Single stimulus acquisition studies have demonstrated that the auditory element appears completely unable to be associated with the same appetitive reinforcer employed in the compound acquisition study. The present experiment, however, suggests that the failure to see a conditioned response is not a failure of association, but is instead a failure of this association to be manifest in performance.

19. D. Regan and K. I. Beverley. Visual Responses to Changing Size and to Sideways Motion for Different Directions of Motion in Depth: Linearization of Visual Responses.

This paper is concerned with how the visual system computes motion in depth from changes in the size of the retinal image.

The 2-Hz oscillations of our stimulus square's edges were equivalent to the square's oscillating along one of 11 trajectories in depth. All 11 trajectories had the same velocity component (V_z) along a line through the eye (i.e. in the z-direction), but the 11 trajectories had different velocities components (V_x) in the x-direction perpendicular to the z-direction. In separate experiments subjects adapted to each trajectory and we measured threshold elevations for two 2-Hz test oscillations, one equivalent to pure z-direction motion and the other equivalent to pure x-direction motion. To a first approximation threshold elevations for the z-direction test were the same for all trajectories, with the greatest departure from constancy (30%) when two edges of the adapting stimulus were stationary (i.e. equivalent to trajectories that just grazed the eye). Adding an 8-Hz "jitter" oscillation to the 2-Hz adapting stimuli "linearised" the visual response so that threshold elevations were rendered accurately constant ($\pm$5%) for all trajectories tested.

We propose that the changing-size channels precisely compute the algebraic difference between the velocities of opposite edges of the target, thus extracting velocity in the z-direction (V_z) independently of the target's trajectory, so that (with added "jitter") this computation is accurately independent of the x-direction component of motion at least up to $V_x=8V_z$. We have no evidence for a complementary channel that computes V_x independently of V_z over any comparable range of $V_x:V_z$ ratios.

20. J. C. Fentress. Motor Traps in Ontogeny.

A characteristic of early behavior in mammals is the frequency of abrupt transitions between functionally diverse activities. Recent observations in rodents, canids, and children indicate that many of these transitions can be accounted for by similarities of movement used in functionally different contexts. These observations reduce interpreted uncertainty of temporally sequenced actions, and provide independent evidence for a developmental hierarchy in integrated performance. They also bear upon how we think about "pieces" of behavior as distinct from their relations in time.

21. G. J. Hollway and M. Ozier. The Encoding of Frequency of Occurrence and the Episodic Memory Trace.

22. B. Rusak. Neural and Hormonal Regulation of Seasonal Reproductive Cycles.

Golden hamsters (and most other animals) undergo seasonal cycles of reproductive activity and quiescence. For hamsters, the external trigger for this cycle is exposure to the shortening daylengths of autumn; the internal neuroendocrine transducer of this signal appears to be the pineal gland. Pineal removal locks the hamster into the reproductive phase regardless of daylength. Destruction of the suprachiasmatic nuclei (SCN) of the hypothalamus interferes with pineal hormonal activity and like pinealectomy prevents seasonal gonadal regression. The original assumption that the effects of SCN on the gonads are mediated solely by inactivation of the pineal have been shown to be wrong. These lesions also appear to affect the neural control of pituitary hormone release, especially prolactin. Prolactin appears to exert a protective effect on the gonads; artificially lowering prolactin levels in SCN-lesioned hamsters can permit short photoperiods to produce gonadal regression. These results suggest that blocking seasonal reductions in prolactin may be a more important action of SCN lesions that inactivation of the pineal gland.